

Abstract

This thesis investigates original research in modelling nonlinear financial time series to predict market regime changes. This is motivated by the postulation that financial series display dynamically changing trends and correlation patterns whose behaviors are influenced by changing regimes, understood as clusters of relatively persistent equilibria of economic dynamics. The changes between periods of distinct characteristic behaviors of time series are often referred as regime switching (Hamilton, 2016). This thesis produces a machine learning binary classification model that predicts the occurrence of a “crisis” regime in financial markets on each day when the average return in the next interval of fixed length would be below the average of all such intervals.

Regime switching models have traditionally been developed in econometrics. This thesis formally defines and investigates the use of machine learning algorithms to model regime changes in finance. Machine learning algorithms are proposed in this investigation for their non-parametric nature, which require no explicit knowledge or assumption on the data generating processes of a dataset such as financial markets with complex cross variable relations and exogeneous influences.

The thesis comprises three investigations, believed by the author to be original contributions in nonlinear financial modelling. The research structure is presented as follows and will be discussed in more detail in section **Error! Reference source not found.**:

Experiment 1: Error! Reference source not found.. This study makes a formal hypothesis and proposal of the generalized occurrences of regime switches in markets. The work presents a theoretical framework of regime switching manifested as nonlinear economic changes on financial variables. The discussions formally define a classification problem for machine learning algorithms to model similar notions of recurrent nonlinear changes in financial markets, as machine learning models are explained to more suitably model short-term market changes.

Experiment 2: Error! Reference source not found.. The study introduces the construction of one regime model by showing regime changes in the S&P 500 index, according to a specific and quantified definition. The discussion presents market data and their derived properties under each regime, whereby switches between regimes give rise to different behavioural characteristics. The insights are used to develop classification models that make inferences on real-time regime changes, as will be shown in investigation 3.

Experiment 3: Error! Reference source not found.. The last investigation presents machine learning algorithms and data science concepts that are utilized to construct regime classification models and evaluate their performances. This work is an illustration of one method to classify real-time regime changes. The discussions should be valuable by showing a generalizable framework to develop a capable model that recognizes regime effects. The results should lead to insights and motivations for further studies and enhancements on the topic.

The thesis presents the following original contributions to sciences, discussed further in section **Error! Reference source not found.**:

1. **Reframe regime switching as a machine learning classification problem.** Nonlinear financial models are scarcely studied, due to the market’s volatile and unpredictable nature that is difficult to model. This work examines the problem by constructing a rigorous yet still practical definition

of market regimes that one could devise methods to identify and examine their occurrences in the market.

2. **Explainable market model through analysis on effective indicators for regime changes.** The study uses variables with clear relation to regime changes, found by exploratory analysis and justified by visualizing their behaviours in different regimes. The variables' indicative power on regime behaviours are validated by being used as source for machine learning algorithms to generate inferences on the state switches.
3. **Lead to generalized regime-switching hypothesis on markets.** An important observation from this study is the general occurrence of regime switching across markets, shown by the development of models that infer regime changes on individual market series by examining changes in collective market behaviours. The observation is naturally expected of markets that are hypothesized to be under dynamic equilibrium.

A paper based on which research has been drafted and will be submitted for publication.